Figure 12.8 SIV mode uses the MAC authentication tag as the IV for encryption. This ensures that the IV will only repeat if the message is identical, eliminating nonce reuse issues that can cause catastrophic security failures. AES-SIV is particularly suited to IoT environments because it only needs an AES encryption circuit to perform all operations (even decryption).

The encryption process works by making two passes over the input:

1. First, a MAC is computed over the plaintext input and any associated data.[9] The MAC tag is known as the Synthetic IV, or SIV.
2. Then the plaintext is encrypted using a different key using the MAC tag from step 1 as the nonce.

The security properties of the MAC ensure that it is extremely unlikely that two different messages will result in the same MAC tag, and so this ensures that the same nonce is not reused with two different messages. The SIV is sent along with the message, just as a normal MAC tag would be. Decryption works in reverse: first the ciphertext is decrypted using the SIV, and then the correct MAC tag is computed and compared with the SIV. If the tags don't match, then the message is rejected.

> **WARNING** Because the authentication tag can be validated only after the message has been decrypted you should be careful not to process any decrypted